

Aymeric Barbin

Ph.D. in Artificial Intelligence

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Lyon, France



Professional Experience

Postdoctoral researcher (*Université de Tampere, Finlande*) 2025–2026

- Developed a data-processing pipeline for high-frequency limit-order-book data using the Deutsche Börse A7 platform
- Extended and evaluated a generative deep-learning architecture (MarS) for limit-order-book modelling and financial time-series prediction

Doctoral Researcher in Artificial Intelligence (*Sapienza University of Rome, Italy*) 2022–2025

- Development of R-Latplan, an architecture for learning reliable and interpretable PDDL action models from images, and evaluation of its robustness under noisy and incomplete observations.
- Integration of logical constraints into neural training through differentiable constraints expressed in fuzzy logic.
- Development of RC-Latplan, a logically equivalent version of R-Latplan with a more compact action model obtained through hierarchical clustering.
- Design and implementation of benchmarks and experimental protocols to evaluate planning performance, coverage, and model robustness.
- Contribution to Visual Reward Machines, a neurosymbolic model for learning finite-state automata from images.

IA / Web developer (*Freelance*) 2018–2023

- AI development, including a computer-vision system for bowling lanes using a Raspberry Pi-mounted camera and OpenCV/YOLO.
- E-commerce development on PrestaShop and Shopify.

Full Stack Web developer (*Ozon Digital*) 2019–2020

Front-end and back-end development for e-commerce websites.

Cybersecurity Engineer (*La Poste & HR Team Group*) 2016–2017

Penetration testing and code and security audits for web applications.

Private Mathematics Tutor (*Freelance*) 2010–2016

Tutoring for high school and bachelor's-level students.

Education

Ph.D. in Artificial Intelligence 2022–2025

Sapienza University of Rome

Thesis: Learning Symbolic Planning Models from Images

Grade: Very Good

Master's Degree in Artificial Intelligence 2020–2021

KU Leuven, Belgium

Thesis: Learning to learn: automatic design of neural network architectures for biomedical signal processing tasks.

Master's Degree in Computer Science — Security, Reliability and Performance of Digital Systems

2015–2016

Pierre and Marie Curie University (Paris 6), France

Bachelor's Degree in Applied Mathematics

2012–2013

Bordeaux Segalen University, France

Compétences IA (liste non-exhaustive)

IA symbolique

Planification PDDL/STRIPS, automates finis, programmation logique et par contraintes, systèmes multi-agents, représentation des connaissances

IA connexionniste

Réseaux de neurones, deep learning, Transformers, CNN/RNN, VAE et VQ-VAE, modèles génératifs

Méthodes d'entraînement et d'évaluation

Apprentissage supervisé/non supervisé/par renforcement, optimisation d'hyperparamètres, régularisation par contraintes logiques, arbres de décision et clustering, analyses statistiques

Langages, outils

Pytorch, Tensorflow/Keras, scikit-learn, weights & biases

Présentations aux Conférences

- Italian Workshop on Planning and Scheduling (IPS-2022, 10th edition), Udine, Italie
- NeSy 2023, 17th International Workshop on Neural-Symbolic Learning and Reasoning, Sienne, Italie
- 36th IEEE International Conference on Tools with Artificial Intelligence, ICTAI 2024, Herndon, USA

Publications

- Barbin, A., Cerutti, F., & Gerevini, A. E. (2024). Learning reliable pddl models for classical planning from visual data. *36th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*
- Umili, E., Argenziano, F., Barbin, A., Capobianco, R., et al. (2023). Visual reward machines. *NeSy 2023, 17th International Workshop on Neural-Symbolic Learning and Reasoning*, 3432, 255–267
- Barbin, A., Cerutti, F., Gerevini, A. E., et al. (2022). Addressing the symbol grounding problem with constraints in neuro-symbolic planning. *Italian Workshop on Planning and Scheduling (IPS-2022, 10th edition)*, 3345

References

Prof. Juho Kanninen

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Prof. Alfonso Emilio Gerevini (superviseur principal)

Full Professor, University of Brescia, Italy
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Prof. Federico Cerutti (co-superviseur)

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